DESIGN AND OPTIMIZATION OF FSAE WHEEL ASSEMBLY COMPONENTS

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Abstract: In Formula Student (FSAE) vehicles, the wheel assembly plays a critical role in handling and performance. This paper aims to optimize the design of the hub and upright for a 10-inch wheel, focusing on reducing weight without compromising structural integrity. Key parameters considered include safety deformation limits, fatigue resistance, and accurate bearing selection to ensure proper fit and function. The design integrates variable camber adjustability for better handling characteristics. SolidWorks was used for 3D modeling, while ANSYS enabled structural and fatigue analysis under various load conditions. Materials were selected based on strength-to-weight ratio and manufacturability. The optimized design resulted in a 15% weight reduction and improved stress distribution, ensuring durability under dynamic loading. The study demonstrates an effective approach to lightweight and high-performance wheel assembly design tailored for FSAE applications
Keywords: FSAE, Wheel Hub, Uprights.

1. INTRODUCTION

Formula SAE is a student-level design competition where teams from various colleges from around the world participate to compete in static and dynamic events. The **wheel assembly** forms the vital interface between the vehicle and the road. More than just a rolling element, it is a symphony of precisely engineered components—**hub, upright, bearing, brake system, and fasteners**—working in unison to translate driver intent into controlled motion. Each part of the assembly must be meticulously designed to withstand extreme forces during acceleration, braking, and cornering, all while contributing to a lightweight and responsive chassis. Our design focuses on achieving the **optimal balance between strength, weight, manufacturability, and serviceability**, ensuring that the wheel assembly not only meets dynamic load requirements but also aligns with the overarching.

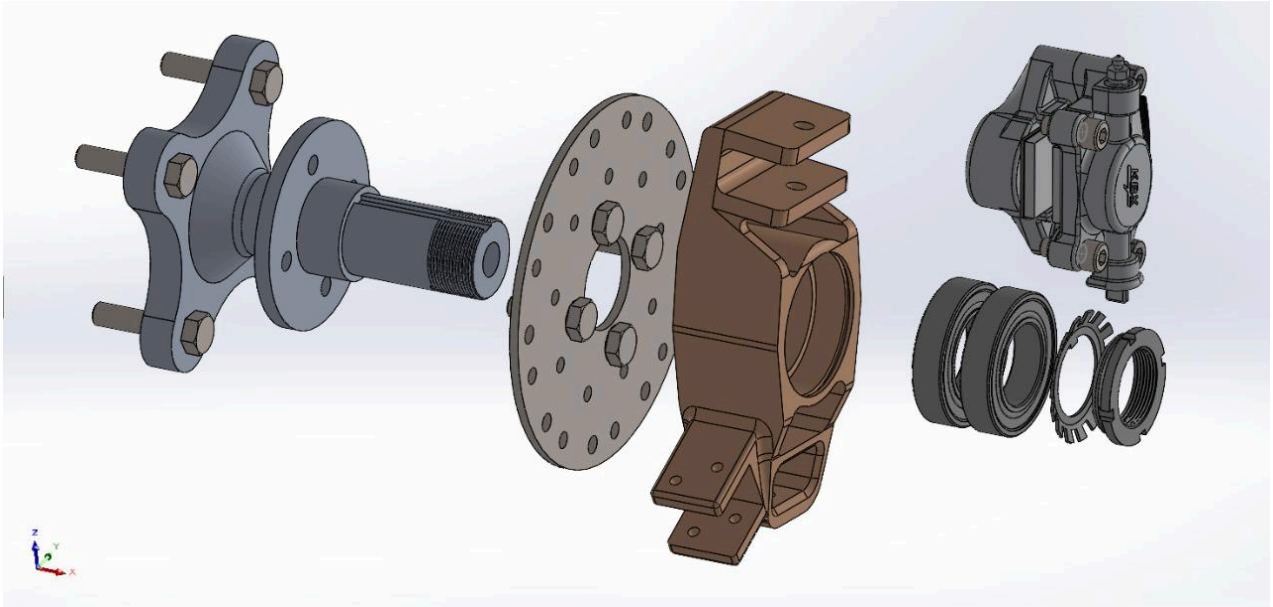


Fig. 1 Wheel Assembly Layout

2. BEARING SELECTION

It is necessary to select the bearing before starting the design of hub and knuckle, as some dimensions are dependent upon size and diameter of the bearing. It serves as the main connection between the rotating hub and the fixed knuckle, allowing smooth wheel rotation while handling significant radial and axial loads generated during cornering, braking, and acceleration. For front and rear wheel assembly, deep groove ball bearings are used in back-to-back arrangements. For our application, SKF 6007 deep groove ball bearing is used in front and rear wheel assembly.

TABLE 1
Bearing Properties and Force Calculations[a]

Designation	6007	Maximum Load on Hub	1.748 kN
Bore Diameter	35 mm	Axial Load on bearing(F_A)	0.828 kN
Outside Diameter	62 mm	Radial Load on bearing(F_R)	2.447 kN
Width	14 mm	L10	274.34 Million Revolutions
Dynamic Load Rating	15.9 kN	Average Load on Hub	0.55 kN
Static Load Rating	10.2 kN	Life at Average Loading	4572.337 Hours
Mass	0.15 kg	F_A/F_R extreme load case	5.91

3. HUBS

3.1 INTRODUCTION:

In a Formula Student (FSAE) vehicle, the **wheel hub** is a critical component that serves as the interface between the suspension system and the wheels. It is responsible for transmitting forces from the wheels to the uprights and ultimately to the chassis, while also supporting the mounting of the brake disc and wheel bearings. Hubs must be precisely designed to withstand high loads under acceleration, braking, and cornering while maintaining minimal weight to support the vehicle's overall performance. Given the emphasis on performance, cost, and manufacturability in FSAE, hubs are often custom-designed using lightweight materials like aluminium alloys and optimized through finite element analysis (FEA). Their design directly influences factors such as **rotational inertia**, **unsprung mass**, and **reliability**—all of which are vital in a competitive racing environment.

3.2 DESIGN PROCEDURE:

Factors to be considered:

a) PCD(Pitch Circle Diameter) of wheel mounting plate
b) Bearing Used
c) PCD of brake rotors
d) Locking methods of bearing
e) Front of rear hub
f) Size of brake calliper

The hub design begins after selecting the wheels, specifically based on the number of lug bolts and the PCD (Pitch Circle Diameter) of the wheels. Once that's fixed, the brake rotor design and its PCD are considered to define the rotor mounting on the hub. The hub's shoulder length depends on the width of the bearing and the locking method used, while the shoulder diameter is determined by the inner diameter (ID) of the selected bearing. For front hubs, the design is relatively simple, focusing only on the bearing seat, wheel mount, and brake disc interface. In contrast, rear hubs are more complex as they must accommodate drivetrain connections like splines or flanges for the driveshaft or CV joint.

3.3 LOCKING METHOD:

To securely retain the bearing on the hub, a **lock nut and tab washer** system has been used. The tab washer is placed between the bearing and the lock nut, with one of its tabs engaged in a keyway or slot on the hub shaft to prevent rotation. After tightening the lock nut to the required preload, one of the washer's tabs is bent into a notch on the nut, effectively locking it in place and preventing loosening due to vibrations or rotational forces. This method provides a reliable and compact solution for axial retention of the bearing, ensuring consistent performance under dynamic loading conditions.

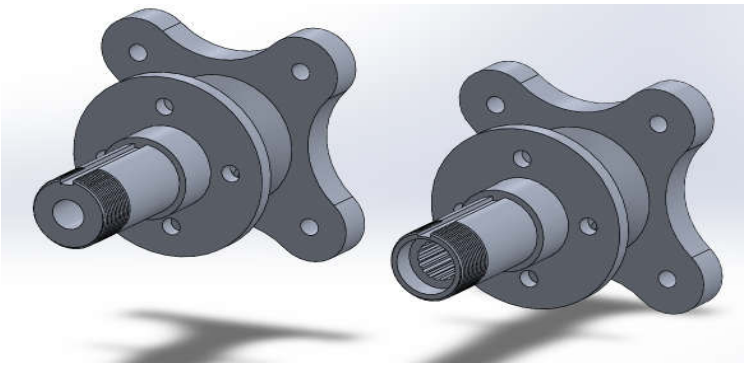


Fig. 2 Front Wheel Hub and Rear Wheel Hub



Fig. 3 Lock Nut and Tab Washer

3.4 MATERIAL SELECTION:

TABLE-2
Properties of Al7075-T6[1]

Young's Modulus (GPa)	71.7
Poisson's Ratio	0.33
Density (kg/m³)	2810
Tensile Strength (MPa)	572
Yield Strength (MPa)	503
Fatigue Strength (MPa)	159

3.5 FORCE CALCULATION AND ANALYSIS:

The hub must have to withstand following forces:

a) Acceleration and Deceleration
b) Force due to Braking
c) Forces at the Cornering

TABLE-3
Data Used in Force Calculation

Parameters	Value	Notations
Mass(Car Plus Driver) (kg)	250	M
Wheel Base (mm)	1550	B
Center of Gravity(CG) Height(mm)	310	H
Track-Width (Front) (mm)	1114	t_f
TrackWidth_Rear (mm)	1093	t_r
Weight Ratio (Front)	0.45	q
Friction Coefficient	1.4	μ
Longitudinal G (braking)	1.5	g_1
Longitudinal G (acceleration)	1	g
Lateral G (Cornering)	1.6	g_2

3.5.1 Force Calculation for front Wheel Hub:

Note: We are considering the peak load case of cornering plus braking simultaneously.

$$\Rightarrow Distance\ from\ rear\ axle\ to\ C.G.(L) = Wheel\ Base \times Weight\ ratio_front$$
$$= B \times q = 1550 \times 0.45 = 697.5\ mm$$
$$\Rightarrow Skid\ Pad\ Diameter = 15.25m \Rightarrow Radius(R) = 7.125\ m$$
$$\Rightarrow Maximum\ Velocity\ while\ turning(V) = 50\ kmph \Rightarrow V = 50 \times (5 \div 18) = 13.89\ m/s$$
$$\Rightarrow Centrifugal\ Load\ transfer\ on\ front\ wheel(P)$$
$$= \left(\frac{Mass \times Velocity}{Radius}^2\right) \times \left(\frac{Center\ of\ Gravity\ Height}{TrackWidth(front)}\right) \times Weight\ ratio$$
$$= \left(\frac{M \times V}{R}^2\right) \times \left(\frac{H}{t_f}\right) \times q = 828.24\ N$$
$$\Rightarrow Dynamic\ Reaction\ Force\ on\ Front\ Wheels\ due\ to\ Longitudinal\ Load\ Transfer(W_{ir})$$
$$= \frac{(Distance\ from\ rear\ axle\ to\ C.G + Longitudinal\ G \times Height\ of\ C.G) \times (Mass \times Acceleration\ due\ to\ gravity)}{2 \times Wheel\ Base}$$
$$= \frac{(L + g_1 \times H) \times (M \times g)}{2 \times B} = 919.6875\ N$$
$$\Rightarrow Total\ Load\ on\ Inner\ Front\ Wheel\ (T) = P + W_{ir} = 1747.93\ N$$
$$\Rightarrow Braking\ Force\ (F_B) = Friction\ Coefficient \times Total\ Load$$
$$= \mu \times T = 2447.11\ N$$

Note: During FEA, we have applied Remote Force at the contact patch of the tire as shown in Fig. 8. Radius of loaded tire=0.229m

TABLE-4
Ansys Simulation Values

Direction of Load	Value	Nature of Load
X	1747.93 N	Static
Y	2447.11 N	Braking (Lateral load transfer)
Z	828.24 N	Cornering (Longitudinal Load transfer)

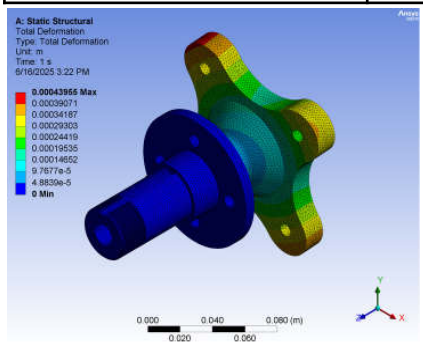


Fig. 5 Total Deformation

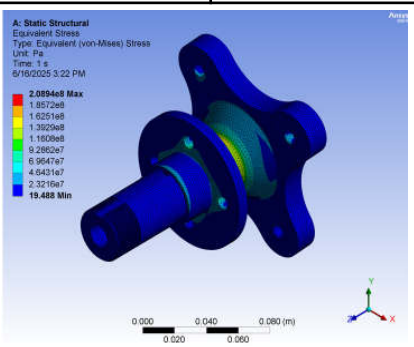


Fig. 6 Von-Mises Stress

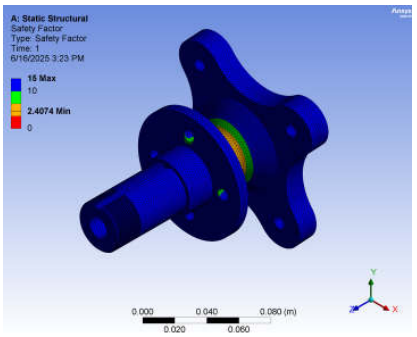


Fig. 7 Factor of Safety

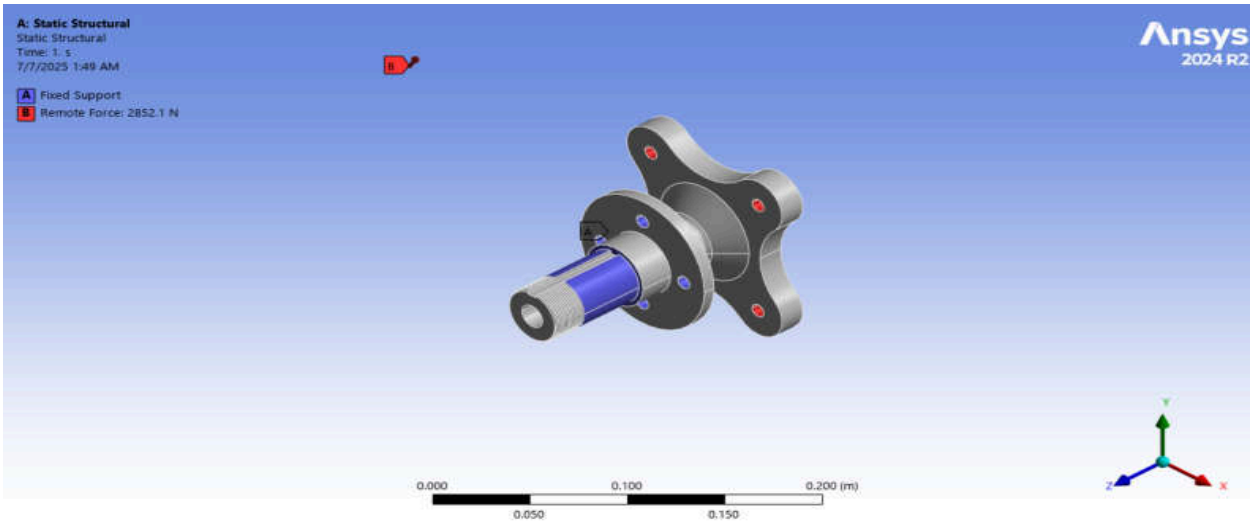


Fig. 8 Location of Forces

3.5.2 Force Calculation for Rear Wheel Hub:

Note: We are considering the peak load case of cornering plus acceleration simultaneously.

$$\Rightarrow Static\ Load\ on\ rear\ wheel(S) = (1 - Weight\ ratio(front)) \times Mass \times g$$
$$= (1 - q) \times M \times g = 674.4375\ N$$

$$\Rightarrow Centrifugal\ force\ while\ turning(C)$$

$$= \left(\frac{Static\ Load}{g}\right) \times \left(\frac{Height\ of\ C.G}{Track\ Width(rear)}\right) \times \left(\frac{Velocity}{Radius}^{2}\right)$$

$$= \left(\frac{S}{g}\right) \times \left(\frac{H}{t_r}\right) \times \left(\frac{V}{R}^{2}\right) = 527.939\ N$$

$$\Rightarrow Dynamic\ Load\ Transfer\ due\ to\ acceleration,\ D_A$$

$$= \frac{Mass \times Height\ of\ C.G. \times Maximum\ Acceleration}{2 \times WheelBase}$$

$$= \frac{M \times H \times a}{2 \times B} = 527.939\ N\ (where\ a = 8 m/s^2)$$

$$\Rightarrow Maximum\ Load\ on\ rear\ wheel = Static\ Load + Centrifugal\ Force$$

$$= S + C = 1202.361\ N$$

$$\Rightarrow Considering\ maximum\ acceleration(a)\ to\ be\ 8\ m/s^2,$$

$$Maximum\ Traction\ Force = Mass \times Maximum\ Acceleration$$

$$= 250 \times 8 = 2000\ N$$

TABLE-5

Ansys Simulation Values

Direction of Load	Value	Nature of load
X	1202.361 N	Static
-Y	2000 N	Traction Force (Lateral)
Z	527.939 N	Cornering Force (Longitudinal)

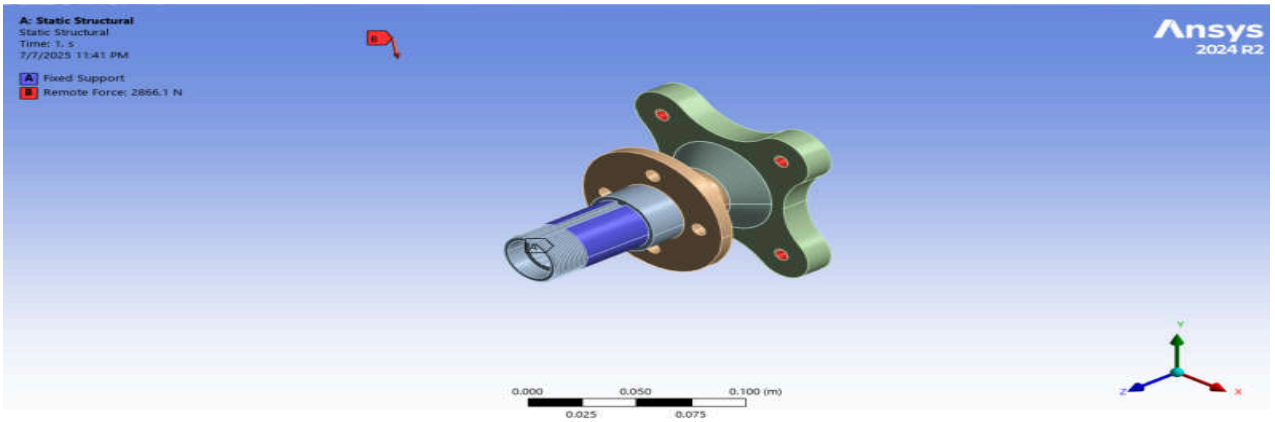


Fig. 9 Location of Forces

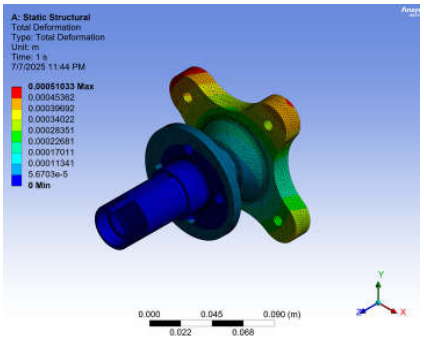


Fig. 10 Total Deformation

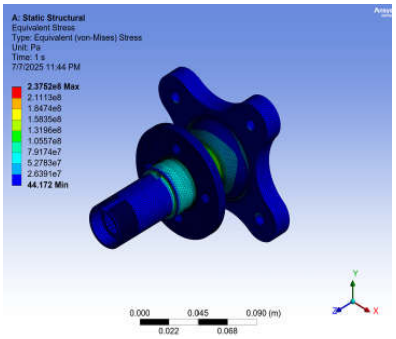


Fig. 11 Von-Mises Stress

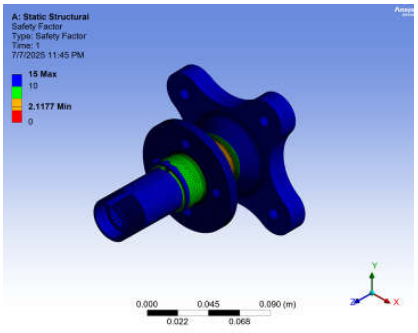


Fig. 12 Factor of Safety

3.6 REQUIRED FATIGUE LIFE:

There are total numbers of three events in which the vehicle should be driven, which are endurance, autocross, and skidpad. The distance covered in each event and in the testing phase was taken into account to determine the required minimum fatigue life.

TABLE 6
Values used for Fatigue Life

Distance for each event	Value(km)
Endurance	22
Autocross	4
Skid pad	2
Tests at event	5
Track test	200
Total	233

Perimeter of the wheel along with the tire : $2 \times \pi \times 216.9\,mm = 1362.8mm = 1.363m$

Required minimum number of cycles = 233000/1.363 = 170946.44= 2×10^{5}

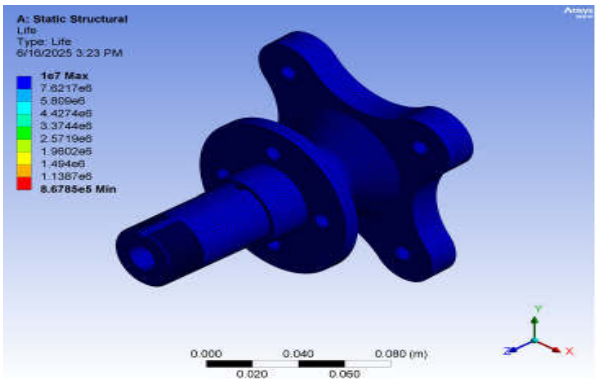


Fig. 12 Fatigue life for front hub

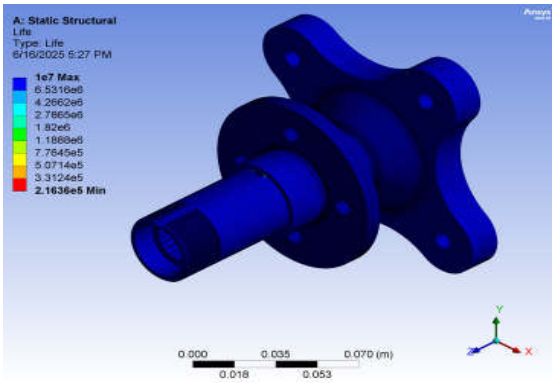


Fig. 13 Fatigue life for rear hub

4. UPRIGHTS

4.1 INTRODUCTION:

In a Formula Student (FSAE) vehicle, the upright is a critical suspension component that connects the wheel hub to the control arms, allowing the transfer of loads between the wheel assembly and the chassis. It supports key elements such as the bearings, brake caliper, and steering arm (in front uprights), while also maintaining precise wheel alignment. Uprights must be lightweight yet strong enough to withstand cornering, braking, and vertical loads, making their design a balance of performance, packaging, and manufacturability. All forces that the car can encounter due to the road and tire interaction can undergo the uprights. The uprights should be sufficiently robust to resist several of those forces occurring at the same time, also as any forces which will happen as a result of a crash or different kind of emergency without failure. Any failure of the uprights would render the automobile un-drivable. Factors that are considered while designing Upright are as follows:

a) Caster Angle

b) Kingpin Axis Inclination
c) Camber Angle
d) Bearings
e) Distance between LBJ and UBJ
f) Brake Calliper
g) Tie Rod mounting

TABLE 7
Data from Vehicle Dynamics

Value	Front Upright	Rear Upright
KPI	6.49°	4°
Caster	2.49°	0°
Distance between UBJ and LBJ	139.78	140.34

4.2 DESIGN PROCEDURE:

- The first step is consideration of KPI and caster angle as per given by the suspension department and from distance of UBJ and LBJ
- Then after designing a bearing house which is dependent on selected bearing and considering tolerance. The space is provided between two bearings for the purpose of ease of assembly and removal and fixes the position of the bearing in the proper manner in the housing.
- The next step after designing of housing the clamps are designed based on KPI and caster angle. After that with the help of a cad model of calliper and mounting dimensions' brake mountings are designed.
- The brackets were designed according to the upright design for UBJ and LBJ.
- The final step of the design of upright is the design of tie-rod mounting and steering arm.

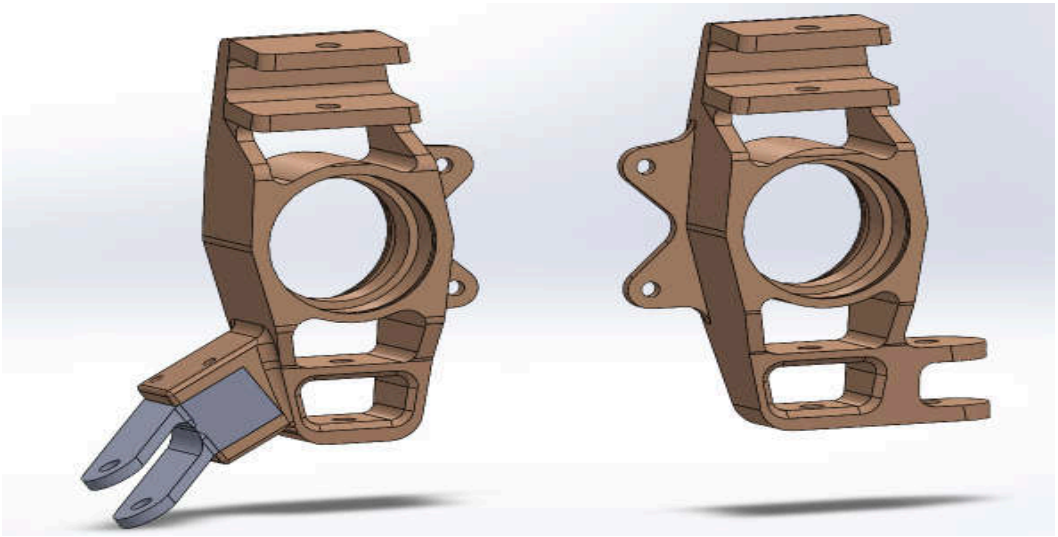


Fig. 14 Front Upright and Rear Upright

4.3 MATERIAL SELECTION:

TABLE 8
Properties of Al6061-T6[h]

Properties	Al6061 T6
Young's Modulus	68.9 GPa
Poisson's ratio	0.33
Density	2700 kg/m^3
Tensile Strength	310 MPa
Yield Strength	276 MPa

4.4 WHY NOT AL7075 T6 FOR UPRIGHTS ALSO:

AL7075-T6 was selected for the wheel hub due to its high strength-to-weight ratio (UTS ~572 MPa, YS ~503 MPa), offering greater safety margins under combined torsional and radial loads. Finite Element Analysis (FEA) confirmed that it performs reliably under worst-case scenarios with minimal deformation and high fatigue resistance, making it suitable for this critical component. For the upright, AL6061-T6 was chosen after FEA validation confirmed that it meets all strength and fatigue requirements under expected loading conditions. Although AL7075-T6 offers higher mechanical properties, the cost increase was not justified for the upright, which experiences relatively lower stress. This choice reflects a performance-cost optimization approach without compromising structural integrity.

4.5 FORCE CALCULATION AND ANALYSIS:

4.5.1 Calculation of forces during maneuvers:

The upright takes longitudinal force during braking and acceleration, and lateral force during cornering. Hence, extreme forces are considered for a situation, when the car brakes during cornering. The total amount of traction is considered constant.

4.5.1.1 Longitudinal Force calculation :

Longitudinal forces acting on the upright originate during phases of vehicle acceleration and deceleration. During braking, a significant deceleration occurs, resulting in dynamic load transfer from the rear to the front axle. This load transfer increases the normal force on the front tires, and the corresponding longitudinal (braking and accelerating) forces are transmitted through the upright.

$$\Rightarrow (Mass\ Transfer) \times 'g' \times WheelBase = (Mass\ of\ vehicle) \times Braking\ g's/Accelerating\ g's \times 'g' \times Height\ of\ C.G.$$

This equation equates the moment generated by the transferred mass with the braking moment about the rear axle, assuming the car decelerates uniformly.

TABLE 9
Braking Force (Front) and Accelerating Force (Rear)

	Units	Front	Rear
Longitudinal Mass transfer	kg	37.5	25
Vertical Load on one wheel	N	919.6875	919.6875
Frictional force on one wheel	N	1287.5625	1287.5625
Braking Force on Upright	N	3462.785735	—
Accelerating Force on Upright	N	—	3462.785735
Diameter of Brake Caliper Mounting Point	mm	6.8	6.8
Distance of hub center to upper A-arm	mm	68	70
Distance of hub center to lower A-arm	mm	72	72
Reaction at upper clamp	N	1440.230625	1419.945687
Reaction at lower clamp	N	-2727.793125	-2707.508187

4.5.1.2 Lateral Force Calculation:

Lateral forces are generated at the upright primarily during cornering maneuvers or steering inputs due to the lateral acceleration experienced by the vehicle. These forces result in mass transfer from the inner to the outer wheels, causing an increase in vertical load on the outer tires and a reduction on the inner ones.

$\Rightarrow (Mass\ Transfer) \times g \times (Track\ Width) = Mass\ of\ Vehicle \times Lateral\ g's \times 'g' \times C.G.\ Height$

TABLE-10
Lateral Force

	Units	Front	Rear
Lateral Mass Transfer	kg	48.94736842	62.39707228
Vertical Load on one wheel	N	1031.986184	1286.552779
Frictional Force	N	1444.780658	1801.173891
Distance of hub centre to upper A-arm	mm	68	70
Distance of hub center to lower A-arm	mm	72	72
Reaction at upper A-arm	N	1616.090364	1986.365009
Reaction at lower A-arm	N	-3060.871022	-3787.5389

4.5.1.3 Force during extreme maneuvers:

Under combined dynamic conditions such as simultaneous braking, acceleration (throttling), and steering, the upright is subjected to the most critical load cases. The most severe loading scenario occurs when the vehicle undergoes braking/accelerating while cornering—a condition where both longitudinal and lateral mass transfers are present simultaneously.

TABLE 11
Maximum Weight Transfer on Wheels

Total weight transfer on front corner	N	86.44736842
Total weight transfer on rear corner	N	87.39707228
Total weight of front corner	N	142.6973684
Total weight of rear corner	N	156.1470723
Coefficient of friction(μ)front		1.4
Coefficient of friction(μ)front		1.4
F_X, front & F_Y, front (Magnitude)	N	1959.805658
F_X, rear & F_Y, rear (Magnitude)	N	2144.523891

4.5.1.4 Linkages forces:

The linkages in a corner of a FSAE race car consists of two wish-bones having four links, a tie/toe rod and a push/pull rod. The tire forces are transferred to the links through upright. Since there are no bending forces, axial forces are developed in all linkages.

The unit vectors are calculated for every link and the tire forces are distributed in the Cartesian plane and directional vector ratios. For calculating the unit vectors of the link, the data have been taken from suspension geometry of a FSAE race car.

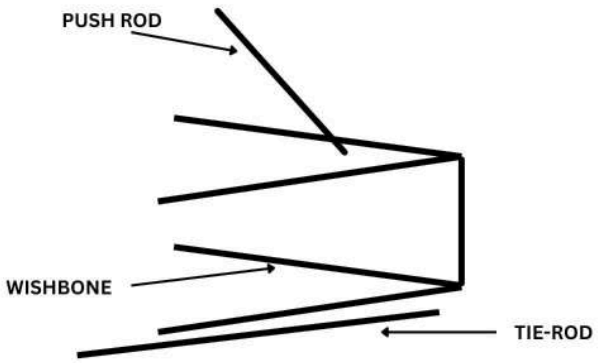


Fig. 15 Linkages of a wheel assembly

TABLE 12
Data from Suspension Geometry

S. No	Points	X (mm)	Y (mm)	Z (mm)	Designation	Length
1	Lower Wishbone front pivot	1700	-270.65	134.34		264.27
2	Lower Wishbone rear pivot	1400	-270.65	132.77		267.63
3	Lower Wishbone outer ball joint	1552.9	-489.71	148.96		
4	Upper Wishbone front pivot	1700	-280	250.9		249.37
5	Upper Wishbone rear pivot	1400	-280	237.6		247.03
6	Upper Wishbone outer ball joint	1546.09	-472.52	288.77		
7	Outer Tie-rod ball joint	1470.01	-452.15	121.35		281.01
8	Inner Tie-rod ball joint	1425	-175	110		
9	Push rod Wishbone end	1546.59	-444.28	330.03		343.08
10	Push rod Rocker end	1550	-282.85	633.2		

TABLE 13
Force Unit Vectors

Unit Vectors	X (mm)	Y (mm)	Z (mm)
3 to 1	0.5566	0.8289	-0.055
3 to 2	-0.5713	0.8185	-0.0604
6 to 4	0.6171	0.7720	-0.1518
6 to 5	-0.5913	0.7793	-0.2071
8 to 7	0.1601	-0.9862	0.0403
10 to 9	-0.0099	-0.4699	-0.8826

The number of links corresponds to the number of columns in the matrix. Solving this matrix yields the axial forces acting within the linkages.

Matrix Array:

$$\begin{matrix} & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$
$$\begin{bmatrix} F_{X1} \\ F_{Y1} \\ F_{Z1} \\ M_{X1} \\ M_{Y1} \\ M_{Z1} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{bmatrix} = \begin{bmatrix} F_X \\ F_Y \\ F_Z \\ M_X \\ M_Y \\ M_Z \end{bmatrix}$$

By inputting the tire forces—derived from force analysis—into the right-hand side (RHS) of the matrix, the axial forces within the suspension linkages can be determined. These computations were carried out using MS Excel, utilizing suspension geometry data for both front and rear assemblies. The corresponding Force-Moment Matrix is formulated for all six linkages as shown below:

TABLE 14
Forces due to Longitudinal Weight Transfer

Location of Forces	Value (in Newtons)
Braking Force on Rear Upright	5767.507781
Braking Force on Front Upright	5270.724393
Reaction at Upper A-arm Front	2192.182614
Reaction at Lower A-arm Front	-4151.988272
Reaction at Upper A-arm Rear	2365.017192
Reaction at Lower A-arm Rear	-4509.541083

TABLE 15
Forces due to Lateral Weight Transfer

Location of Forces	Value (in Newtons)
Reaction at Upper A-arm Front	2237.142857
Reaction at Lower A-arm Front	-4237.142860
Reaction at Upper A-arm Rear	2205.633803
Reaction at Lower A-arm Rear	-4205.633800

TABLE 16
Steering Force and Push Rod Force

Scrub radius	mm	62
Pivot radius of tie rod	mm	123.96
Static Load on one wheel	N	1399.861184
Frictional Force Static	N	1959.805658
Torque about steering axis	N-mm	121507.9508
Force acting on Tie-rod	N	980.2190286
Force acting on Push-Rod	N	920.36

4.6 RESULTS FOR FRONT UPRIGHT:

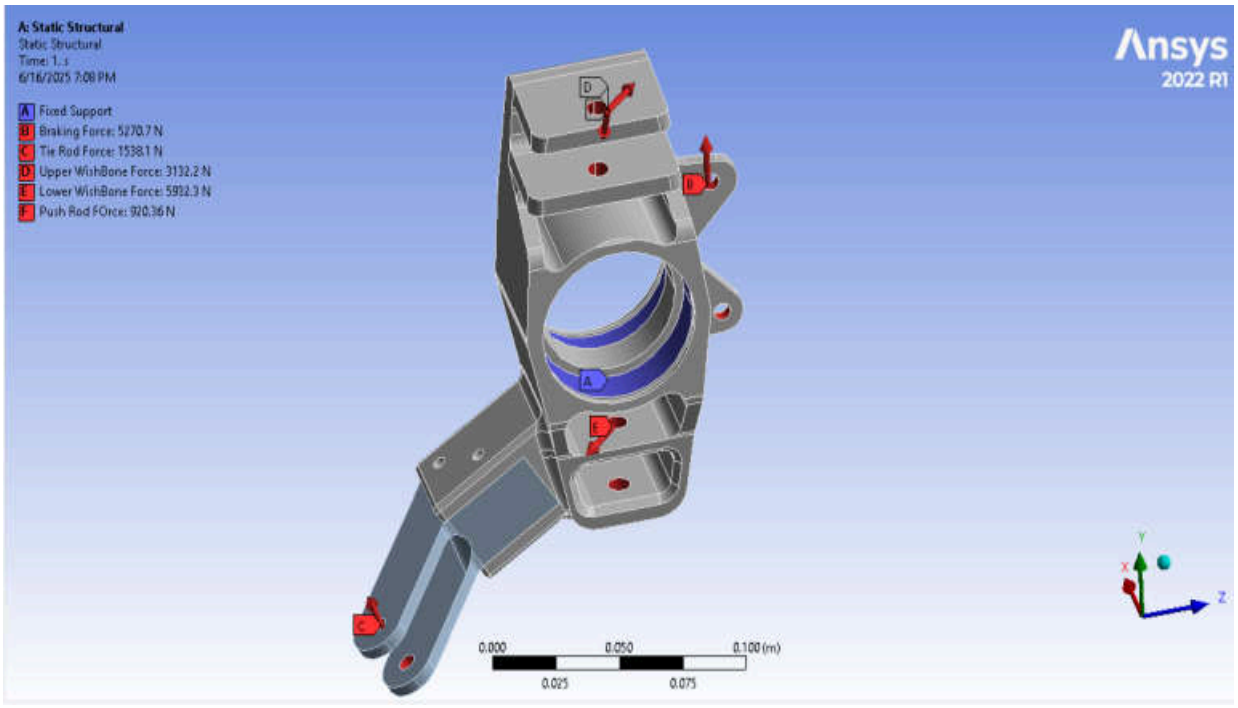


Fig. 15 Boundary Conditions

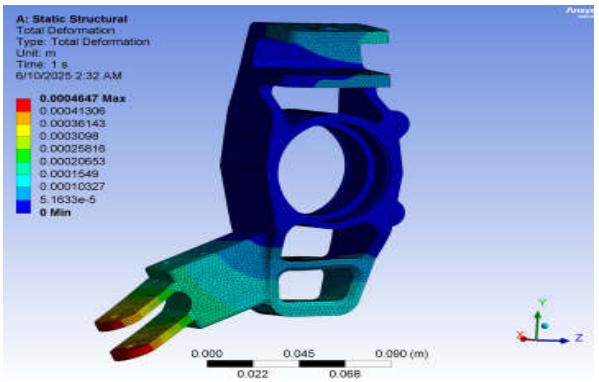


Fig. 16 Total Deformation

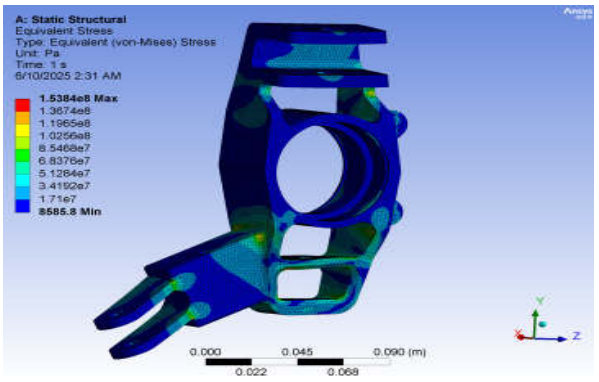


Fig. 17 Von-Mises Stress

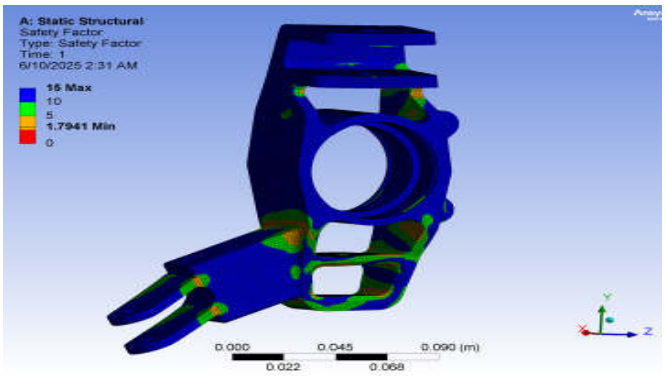


Fig. 18 Factor of Safety

4.7 RESULTS FOR REAR UPRIGHT:

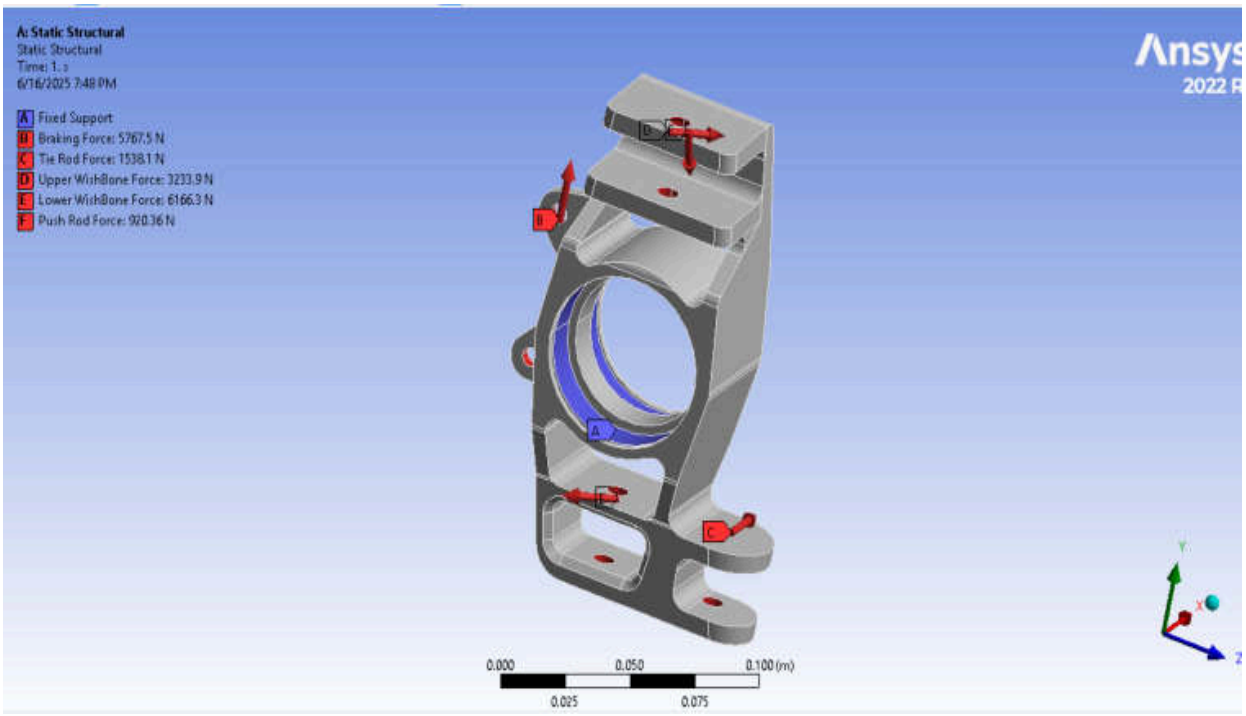


Fig. 19 Boundary Conditions

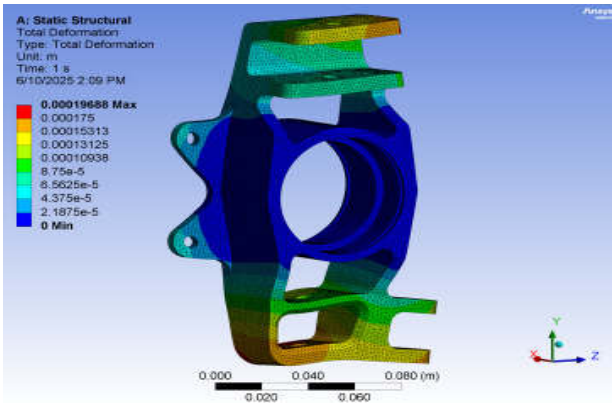


Fig. 20 Total Deformation

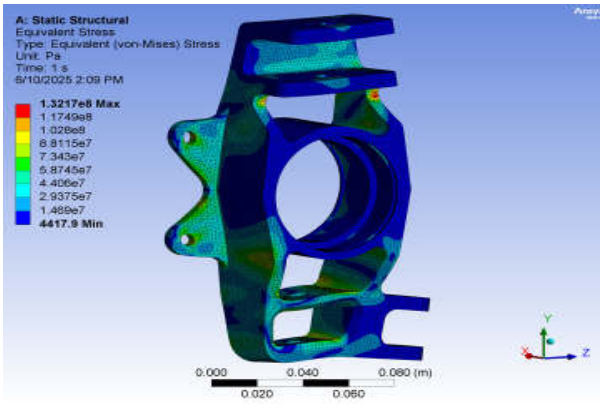


Fig. 21 Von-Mises Stress

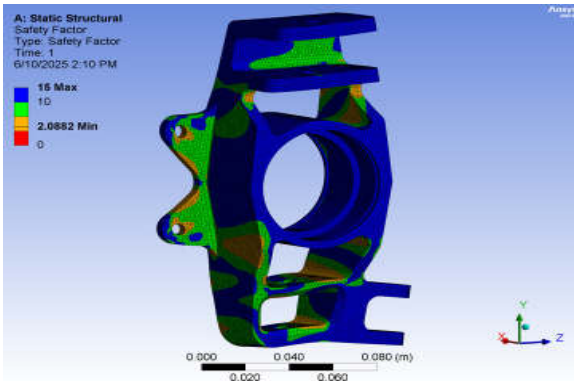


Fig. 22 Factor of Safety

5. CONCLUSIONS

A mathematical model has been generated in MS Excel which can be used for any suspension geometry just by changing the coordinates of the suspension point and other vehicle parameters. From all the analyses and from the results which we obtained, the total deformation is very low and also the stress generated are in the permissible limit of the selected material. In all the analysis the FOS is more than 1.5 so overall the design of the hub and upright is safe.

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